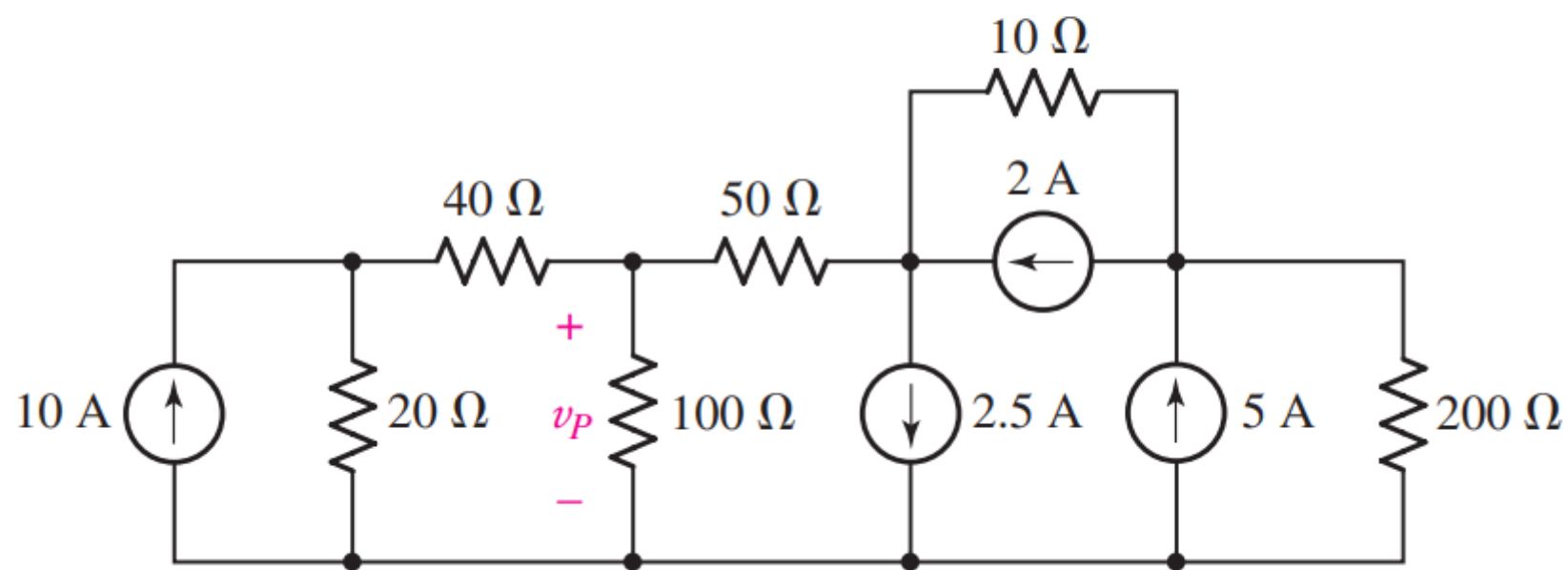


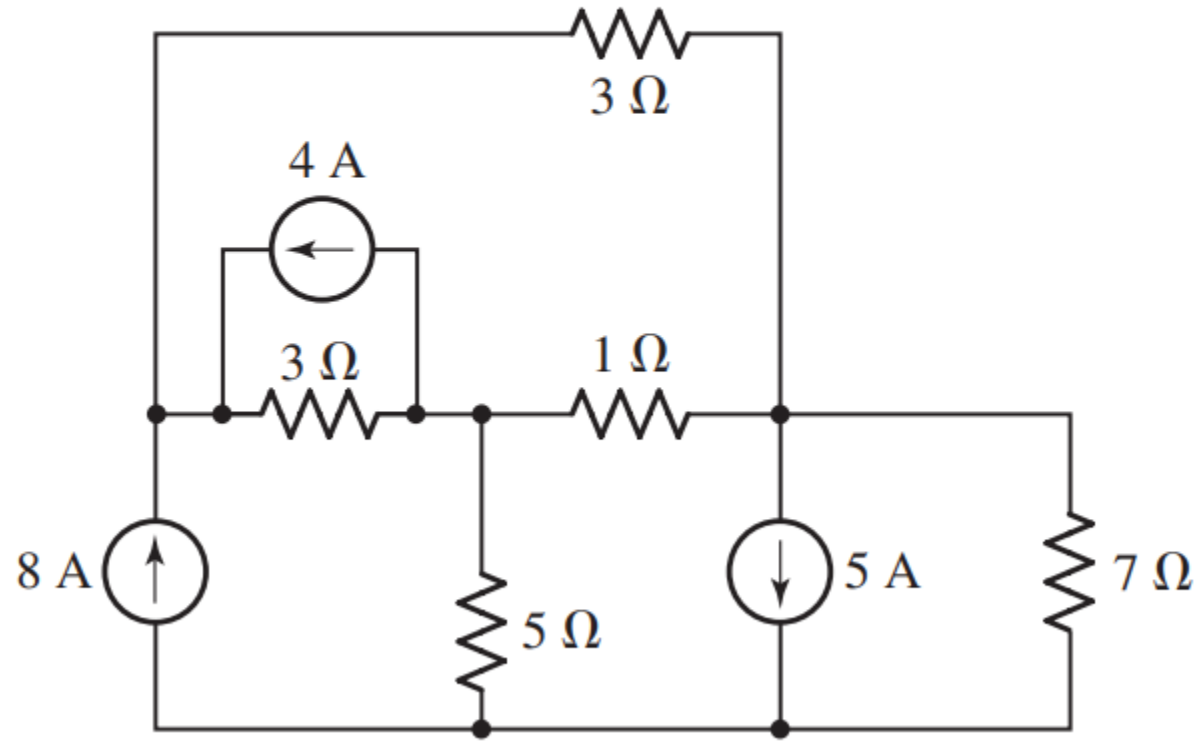
HW6: Node Analysis

12. Use nodal analysis to find v_P in the circuit shown in Fig. 4.38.



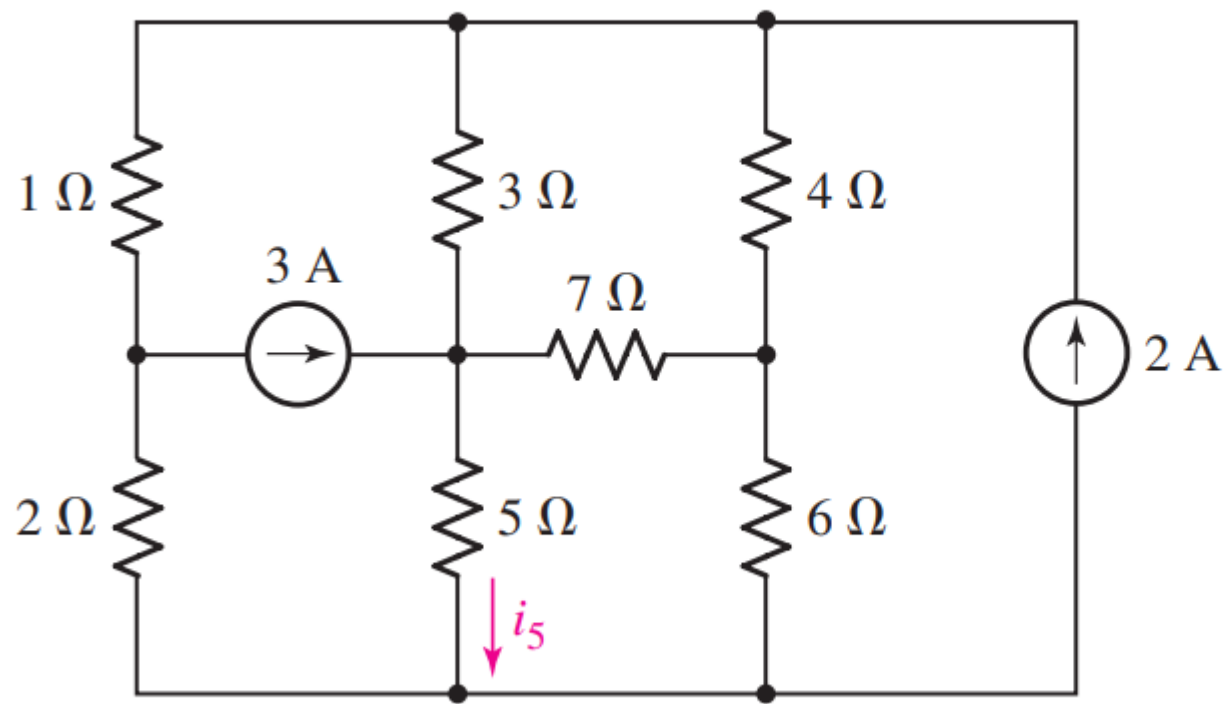
■ FIGURE 4.38

13. Using the bottom node as reference, determine the voltage across the $5\ \Omega$ resistor in the circuit of Fig. 4.39, and calculate the power dissipated by the $7\ \Omega$ resistor.

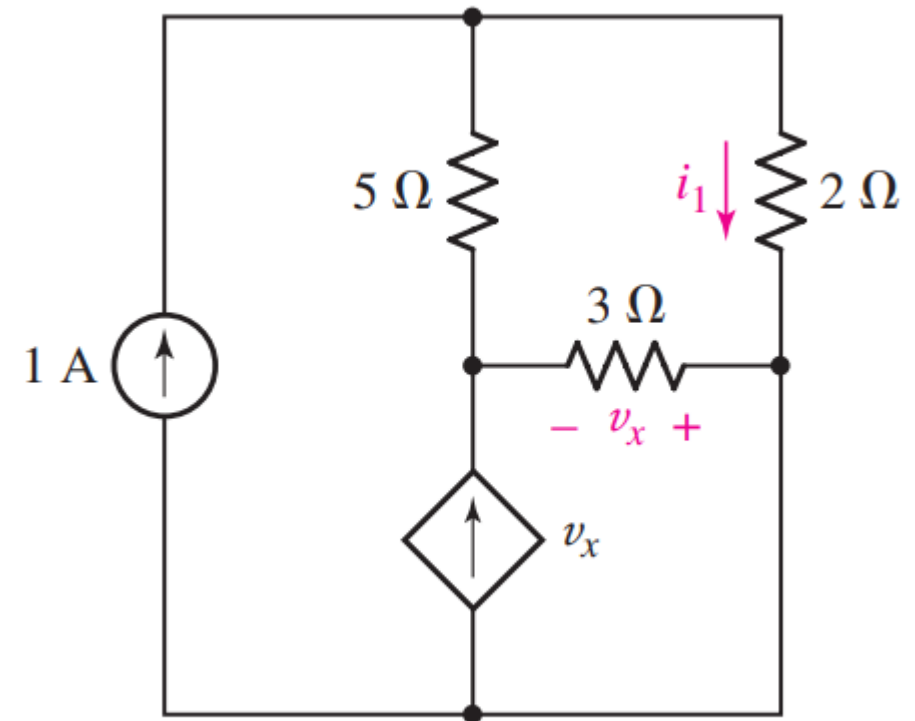


■ FIGURE 4.39

14. For the circuit of Fig. 4.40, use nodal analysis to determine the current i_5 .

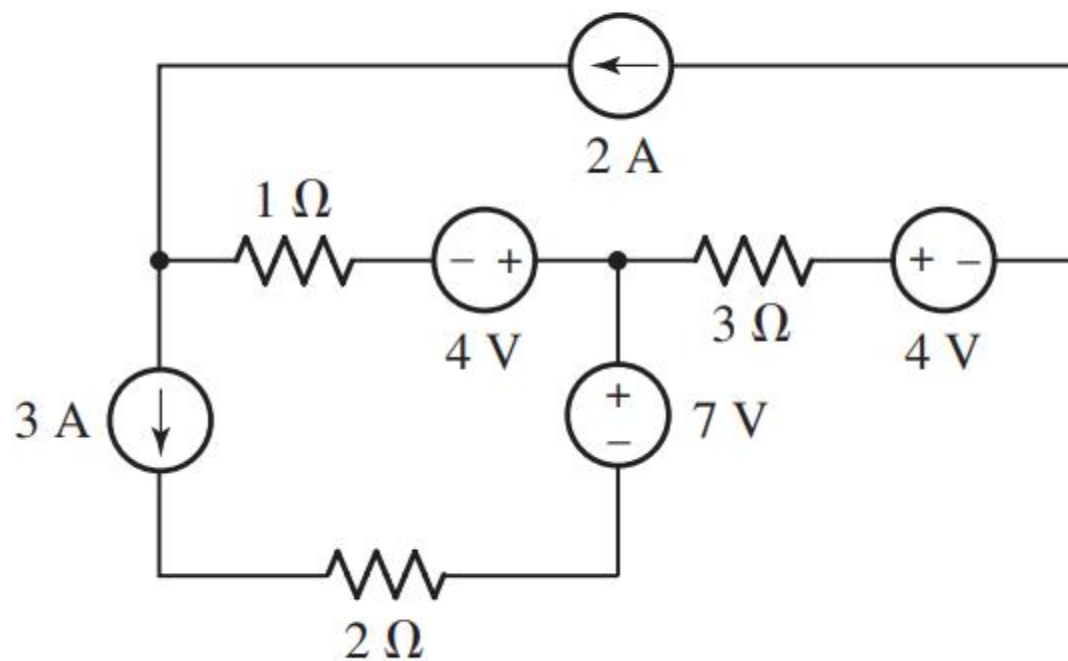


■ FIGURE 4.40



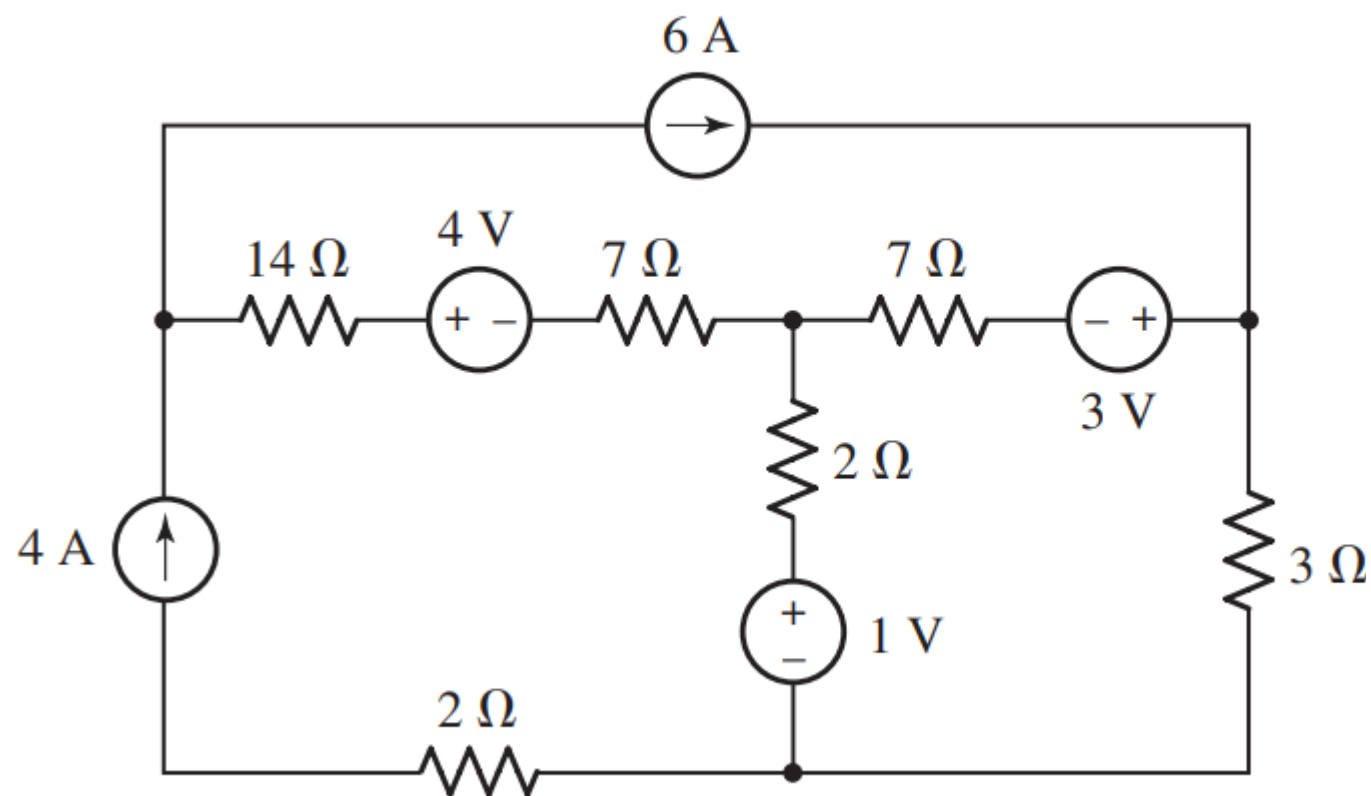
■ FIGURE 4.43

21. Employing supernode/nodal analysis techniques as appropriate, determine the power dissipated by the $1\ \Omega$ resistor in the circuit of Fig. 4.47.

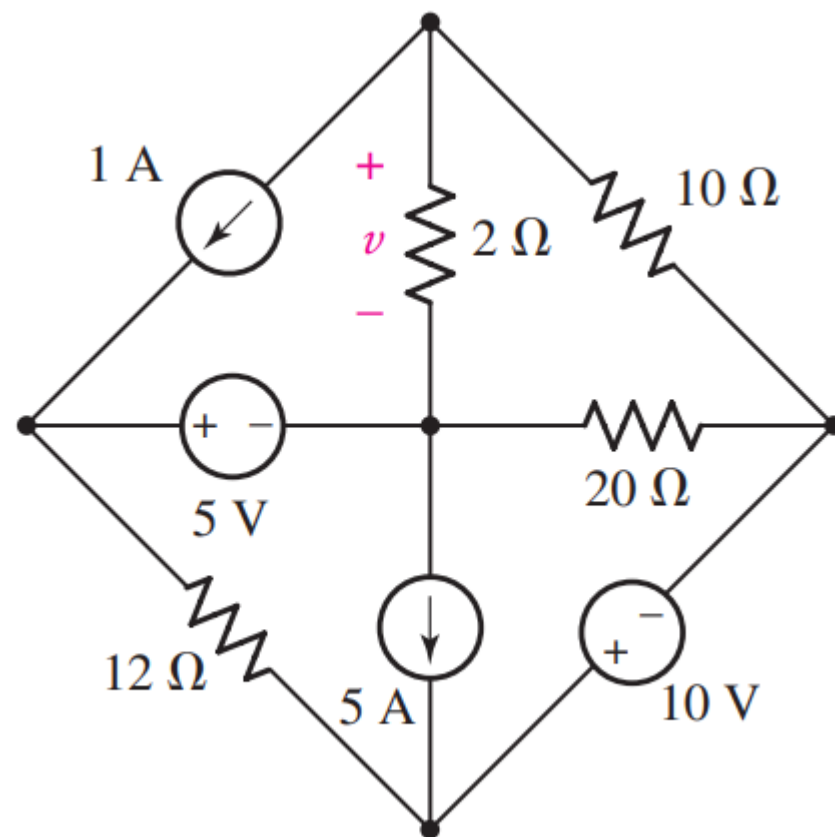


■ **FIGURE 4.47**

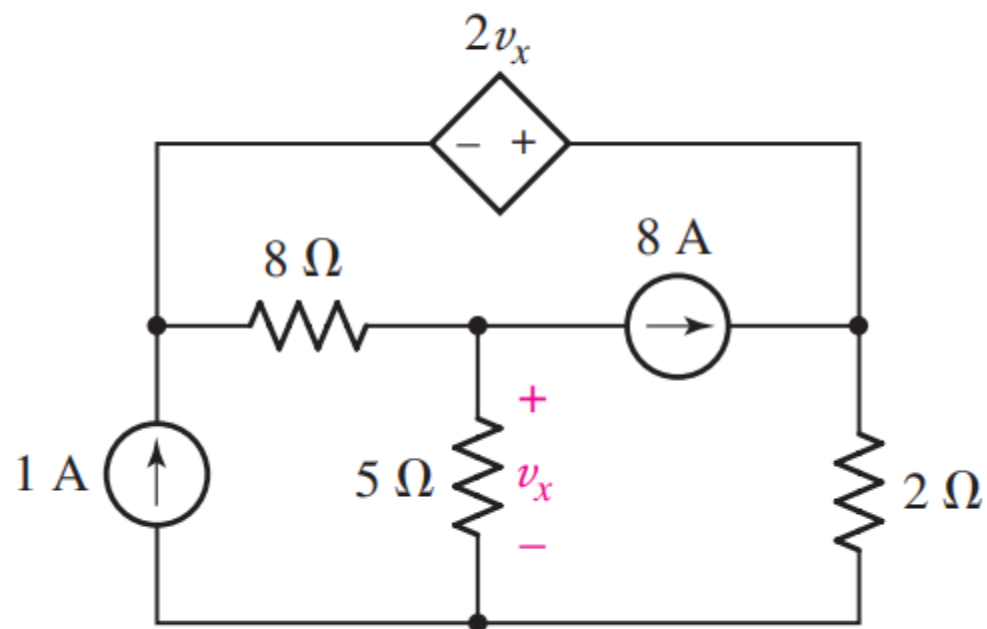
22. Referring to the circuit of Fig. 4.48, obtain a numerical value for the power supplied by the 1 V source.



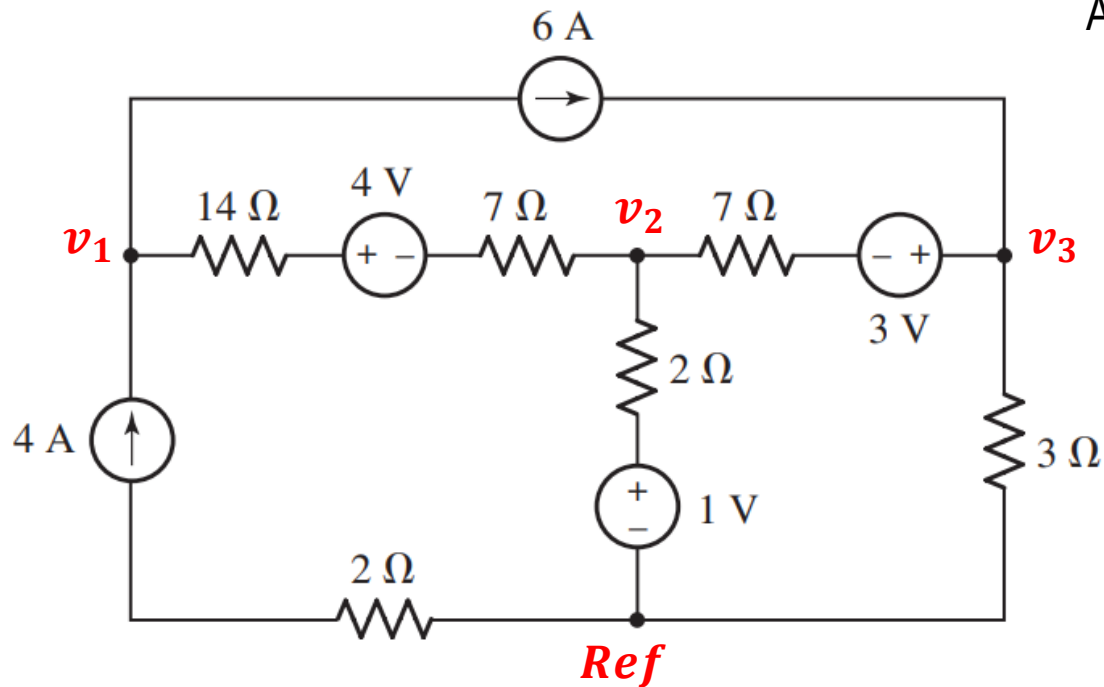
■ FIGURE 4.48



24. Determine the voltage v_x in the circuit of Fig. 4.50, and the power supplied by the 1 A source.



■ **FIGURE 4.50**



Applying KCL at node 2:

$$\frac{v_2 + 4 - v_1}{14 + 7} + \frac{v_2 - 1}{2} + \frac{v_2 + 3 - v_3}{7} = 0$$

$$\frac{v_2 + 4 - v_1}{21} + \frac{v_2 - 1}{2} + \frac{v_2 + 3 - v_3}{7} = 0$$

Applying KCL at node 3:

$$\frac{v_3}{3} + \frac{v_3 - 3 - v_2}{7} = 6$$

Applying KCL at node 1:

$$\frac{v_1 - 4 - v_2}{14 + 7} + 6 = 4$$

$$\frac{v_1 - 4 - v_2}{21} = -2$$

$$v_1 - 4 - v_2 = -42$$

$$v_1 - v_2 = -38$$